

Competitive Exam Practice 7 — Answer Key

Microprogrammed Control Unit Design

PCC CS-401: Computer Organization and Architecture

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- Q1 (D) all of the above.** Horizontal microprogramming dedicates one bit per control signal, so each bit directly *is* the signal (no decoder needed), which is exactly why the microinstruction word is wider than the encoded-field vertical alternative. See the provenance note in the questions file for this answer's verification status.
- Q2 (C).** Vertical encoding groups mutually-exclusive signals into short encoded fields, which must be expanded by a decoder before the datapath sees valid one-hot signals — that decode step is exactly the extra delay vertical encoding pays for its narrower control word.
- Q3 4.** $\lceil \log_2 16 \rceil = 4$ ($2^4 = 16$ exactly).
- Q4 (B) Mapping unit.** The mapping unit converts the opcode in IR into the starting microaddress of that instruction's microprogram and loads it into MPC; control memory only supplies control words *after* MPC already points at the right row.
- Q5 (B).** Hardwired control has no memory-read delay at all (pure gates); horizontal microprogramming reads a wide control word directly with no decode step; vertical microprogramming pays an additional decoder delay on top of the memory read, making it the slowest of the three.
- Q6 11.** $3 + 3 + 5 = 11$.